

# Sentiment Analysis Extension

<https://github.com/aiyshwary/Video-based-and-audio-based-sentiment-analysis>

Python

Hugging Face

NLP

RapidMiner

Audio Processing

## OVERVIEW

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A multimodal sentiment analysis project that independently extracts and analyses sentiment from both the video (visual) and audio streams of a recording, combining insights from two separate NLP and signal processing pipelines.

## IMPACT

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Built a dual-stream sentiment analysis pipeline that separates video and audio from a recording and independently classifies sentiment from each modality, enabling richer multimodal emotional understanding.

## TECHNICAL WALKTHROUGH

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### Sentiment Analysis Extension

#### Problem Statement

Understanding customer opinions and emotional tone from text data is crucial for decision-making in domains like customer feedback, reviews, and social media.

#### Approach

##### Built a sentiment analysis pipeline using:

NLP preprocessing (tokenization, cleaning, normalization) Pre-trained language models (via Hugging Face where applicable)

##### Supports classification into:

Positive

Negative

Neutral sentiments

#### Methodology

##### Text data is cleaned and vectorized

Models are trained or fine-tuned on labeled sentiment datasets

##### Predictions include:

Sentiment label

Confidence score

#### Applications

Customer feedback analysis

Product review monitoring

Social media sentiment tracking

Deployment

**Entire system deployed through AI-Hub using RapidMiner**

**Images and data stored in AWS S3**

**Inference handled via automated workflows**

**Supports:**

**Real-time predictions**

**Error handling**

**Scalable deployment**

**Conclusion**

**This project delivers a complete AI ecosystem for**

**Image Classification**

**Object Detection**

**Sentiment Analysis**

The key strength lies in its flexibility, modularity, and deployability, allowing users to

**Choose models based on requirements**

**Fine-tune on custom datasets**

**Deploy seamlessly in production environments**

It bridges the gap between research-level models and real-world industrial applications.

#### **KEY DETAILS**

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- Extracts audio track from video files and applies speech-to-text followed by NLP-based sentiment classification.
- Processes video frames independently to infer visual sentiment cues from facial expressions and scene context.
- Outputs positive, neutral, or negative sentiment labels with confidence scores per modality.
- Designed for use cases including customer feedback analysis, social media monitoring, and presentation review.
- Implemented using Hugging Face pretrained models and RapidMiner for pipeline orchestration.